

- Q.29 Explain Wheat Stone Bridge.
 Q.30 Explain any five front panel controls of CRO.
 Q.31 Explain Logic Probe.
 Q.32 Explain De-Sauty's bridge.
 Q.33 Explain different methods of Measurement.
 Q.34 What is Standard and explain its different types.
 Q.35 Write any five specification of CRO

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain Digital Storage Oscilloscope (DSO) with block diagram.
 Q.37 Explain construction and working principle of Permanent Magnet Moving Coil (PMMC) instrument.
 Q.38 Write short note on following :
 a) Logic Analyzer
 b) Logic Pulser

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**3rd Sem / Eltx, EI, Med. Eltx, Mechatronics (6th Sem),
 GE(4th Sem), Power Eltx, Elect. & Eltx. Engg.**

Subject:- Electronic Instruments and Measurement

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is caused by careless handling?
 a) Systematic error b) Gross error
 c) Random error d) None of the mentioned
- Q.2 Write full form of DC
 a) Direct coil b) Direct current
 c) Direct cycle d) None of the above
- Q.3 Which one of the following instruments is used only for DC measurement?
 a) PMMC b) Moving Iron Type
 c) Both a and b d) None of the above
- Q.4 What is the full form of CRO
 a) Common Ray oscillator
 b) Combined Ratio Oscilloscope
 c) Cathode Ray Oscilloscope
 d) None of these

- Q.5 _____ provide sinusoidal signal at the output.
- Oscillator
 - Signal Generator
 - DC Tachometer generator
 - None of the above
- Q.6 Which among the following is a part of the Cathode Ray tube?
- Control Electrode
 - Electron Gun
 - Focusing System
 - All of the above
- Q.7 Full form of DSO is
- Direct Signal Operator
 - Digital System Operator
 - Digital Signal Oscillator
 - Digital Storage Oscillator
- Q.8 In the simplest form, AC bridge consist of _____
- arms, source and a detector
 - arms and source
 - Source and detector
 - arms and detector
- Q.9 Q-Meter is used to measure
- Quality factor
 - Power factor
 - Current and voltage
 - None of the above
- Q.10 Logic Probe is used to check
- Logic state of analog circuit
 - Logic state of digital circuit
 - Logic state of analog and digital circuit
 - All of above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is precision?
- Q.12 What is Deflecting Torque.
- Q.13 Expand CRT.
- Q.14 PMMC is used to measure _____. (AC/DC)
- Q.15 What is Time Period.
- Q.16 What is duty cycle?
- Q.17 How many arms are there in a bridge circuit?
- Q.18 What is Distortion Factor Meter.
- Q.19 What is Range.
- Q.20 Expand DVM.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain different types of Error in measurement.
- Q.22 Describe the working principle of Function Generator.
- Q.23 Write any five difference between Analog Multimeter and Digital Multimeter.
- Q.24 Explain CRT with diagram.
- Q.25 Explain working principle of Q-meter.
- Q.26 Explain the working principle of Dual Slope Type Digital Voltmeter.
- Q.27 Describe the working principle of Pulse Generator.
- Q.28 Write any five difference between AC bridge and DC bridge

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